

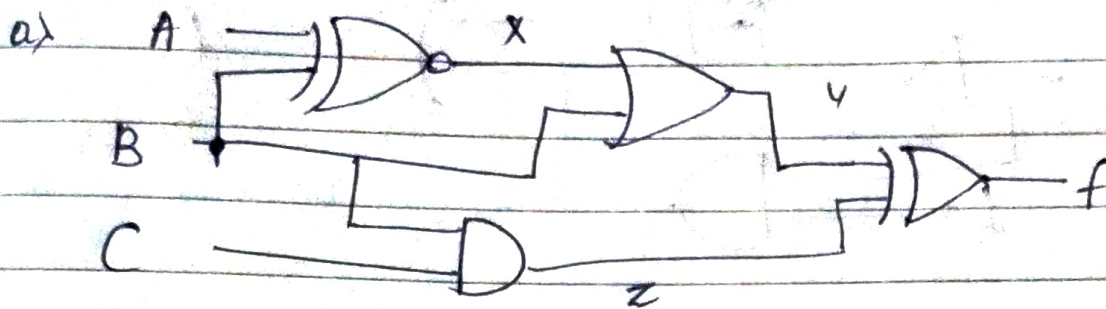
Assignment 2

classmate

Date _____

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Q1 Derive the output expression 'f' for the below ckt & draw the simplified cets.



$$x = A'B' + AB$$

$$y = x + B = (A'B' + AB) + B$$

$$= A'B' + B + AB + B$$

$$y = A' + B$$

$$z = BC$$

$$f = (A' + B) \oplus BC$$

Simplification.

$$x \oplus y = x'y + xy'$$

$$(A' + B)' = AB'$$

$$f = (AB')BC + (A' + B)(BC)'$$

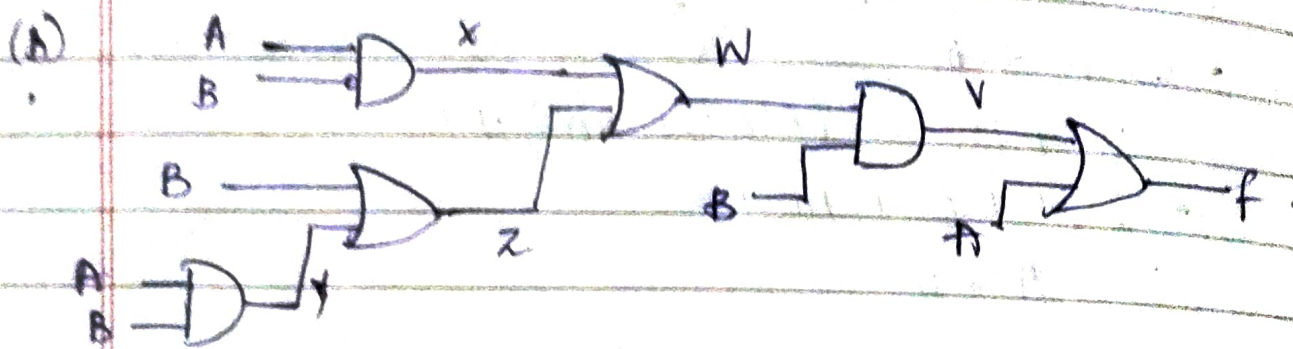
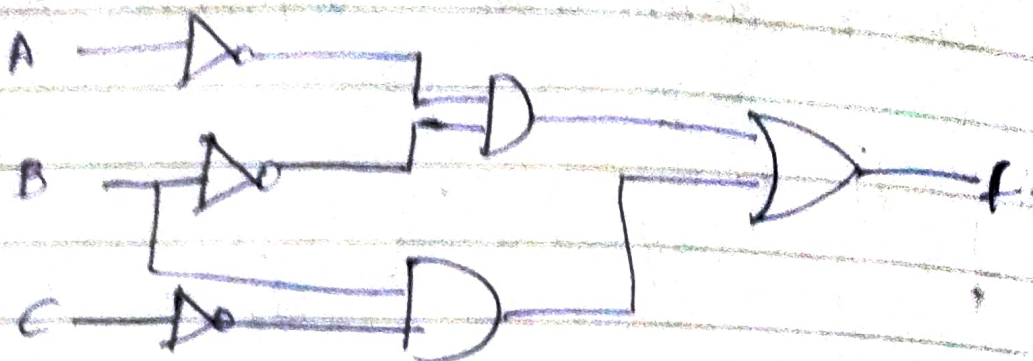
$$\rightarrow AB'BC = 0$$

$$f = (A' + B)(\bar{B} + \bar{C})$$

$$= \bar{A}\bar{B} + \bar{A}\bar{C} + \bar{B}\bar{B} + B\bar{C}$$

$$f = \bar{A}\bar{B} + B\bar{C}$$

simplified ckt



$$\Rightarrow x = A\bar{B}$$

$$y = AB$$

$$z = AB + B$$

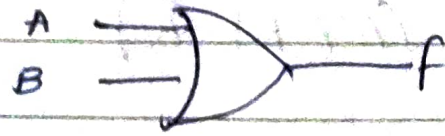
$$w = \bar{A}A\bar{B} + B$$

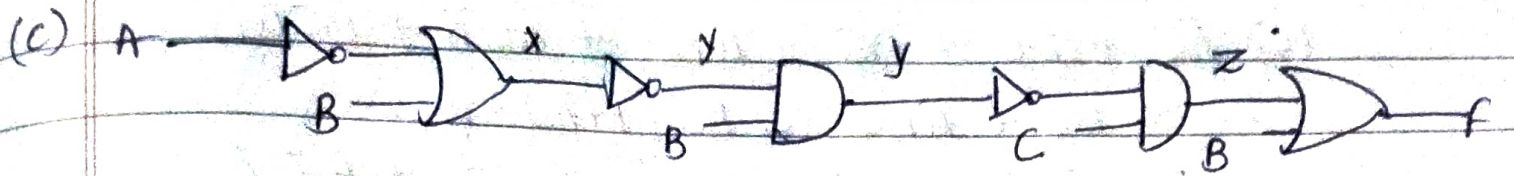
$$z = (A+1)B = B$$

$$N = A + B$$

$$v = B$$

$$f = A + B$$





$$X = \overline{A + B} \quad Y = \overline{\overline{A + B}} = A \cdot B = A \overline{B} B = 0$$

$$Z = 1 \cdot C = C$$

$$f = C + B$$



Q2) Minimize the following boolean expressions using boolean postulates.

a) $\overline{A} C \overline{D} E + \overline{A} \overline{B} \overline{D} + A B C E + A B D$

$$f = E(\overline{A} \overline{D} + A B) + \overline{A} \overline{B} \overline{D} + A B D$$

$$f = C E(\overline{A} \overline{D} + A B) + \overline{A} \overline{B} \overline{D} + A B D$$

this is in its minimum SOP form.

b) $\overline{A} \overline{B} \overline{C} + A B D + \overline{A} C + \overline{A} C \overline{D} + A \overline{C} D + A B \overline{C}$

$$\Rightarrow \overline{A} C + \overline{A} C \overline{D} \Rightarrow \overline{A} C$$

$$F = \overline{A} \overline{B} \overline{C} + A B D + \overline{A} C + A \overline{C} D + A B \overline{C}$$

$$= \overline{A} \overline{B} \overline{C} + A B \overline{C} = \overline{B} \overline{C} (A + \overline{A}) = \overline{B} \overline{C}$$

$$F = \overline{B} \overline{C} + A B D + \overline{A} C + A \overline{C} D$$

$$(c) \bar{A}\bar{C} + DC + A\bar{B} + \bar{A}BC + AC\bar{D} + B\bar{C}\bar{D}$$

$$\Rightarrow BC + \bar{A}BC = BC(1 + \bar{A}) = BC$$

$$F = \bar{A}\bar{C} + A\bar{B} + BC + AC\bar{D} + \bar{B}\bar{C}\bar{D}$$

$$d) \bar{x}_1 \bar{x}_3 \bar{x}_4 + x_3 x_4 + \bar{x}_1 \bar{x}_2 x_4 + x_1 \bar{x}_3 x_4 + x_2$$

$$= x_2 + x_1 x_2 \bar{x}_3 x_4 = x_2 (1 + x_1 \bar{x}_3 x_4) = x_2$$

$$F = \bar{x}_1 \bar{x}_3 \bar{x}_4 + x_3 x_4 + \bar{x}_1 \bar{x}_2 x_4 + x_2$$

$$x_2 + x_2 \bar{x}_1 x_4 \{ (A + A'B) \cdot (A+B) \}$$

$$= x_2 + \bar{x}_1 x_4$$

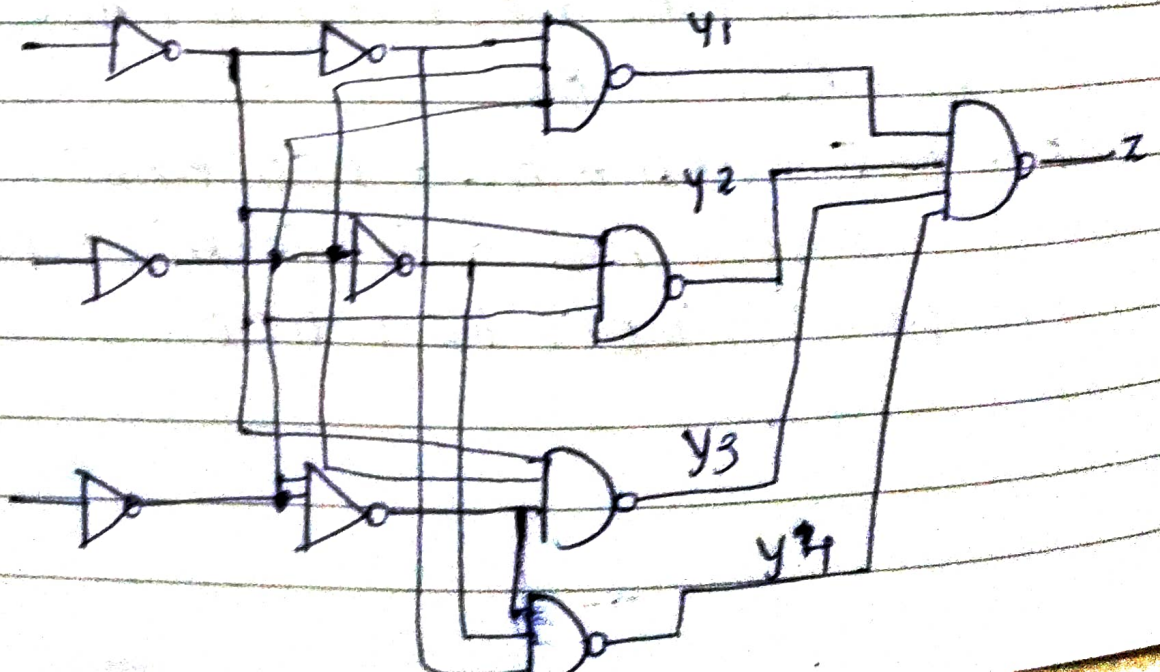
$$F = \bar{x}_1 \bar{x}_3 \bar{x}_4 + x_3 x_4 + x_2 + \bar{x}_1 x_4$$

$$= \bar{x}_1 (\bar{x}_3 \bar{x}_4 + x_4) + x_3 x_4 + x_2$$

$$= \bar{x}_1 (x_4 + \bar{x}_3) + x_3 x_4 + x_2$$

$$F = \bar{x}_1 x_4 + \bar{x}_1 \bar{x}_3 + x_3 x_4 + x_2$$

Q3 Which single gate can represent the functionality of given circuit?



$$y_1 = \overline{A}\overline{B}\overline{C}, \quad y_2 = \overline{A}B\overline{C}, \quad y_3 = \overline{A}B\overline{C}, \quad y_4 = \overline{A}BC$$

$$y_1 = \overline{A} + B + C, \quad y_2 = A + \overline{B} + C, \quad y_3 = A + B + \overline{C}, \quad y_4 = \overline{A} + \overline{B} + \overline{C}$$

$$y = [(\overline{A} + B + C) \cdot (A + \overline{B} + C) \cdot (A + B + \overline{C}) \cdot (\overline{A} + \overline{B} + \overline{C})]'$$

$$= \overline{A}\overline{B}\overline{C} + \overline{A}B\overline{C} + \overline{A}\overline{B}C + ABC$$

$$y = A \oplus B \oplus C$$

∴ It is a xor gate.

Q4. Minimize the following boolean expression using k maps.

$$a) f(a, b, c, d) = \sum_m (4, 6, 8, 10, 11, 12, 15) + \sum_d (3, 5, 7, 9)$$

ab \ cd	00	01	11	10
00	0	0	X	0
01	1	X	X	1
11	1	0	1	0
10	1	X	1	1

$$f = b\overline{c}\overline{d} + a\overline{b} + cd + \overline{a}b$$

b) $f(a, b, c, d) = \prod_m (0, 2, 4, 6, 7, 9), \bar{\pi}_d (10, 11)$

ab \ cd	00	01	11	10
00	0	1	1	0
01	0	1	0	0
11	1	1	1	1
10	1	0	x	x

$$f = (a+d) (a+\bar{b}+\bar{c}) (\bar{a}+b+\bar{d})$$

$$f = (a+d) (a+\bar{b}+\bar{c}) (\bar{a}+b+d)$$